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SERVICE INFORMATION

GENERAL

WARNING

Gasoline is extremely flammable and is explosive under certain conditions. Work in a well ventilated area. Do not smoke or allow flames or sparks in the work area.

- The front cylinders use down draft carburetors.
- When disassembling fuel system parts, note the locations of the O-rings. Replace them with new ones on reassembly.
- The float bowls have drain screws that can be loosened to drain residual gasoline.
- Fuel pump inspection is in section 21.
- The No. 1 and No. 3 carburetors use different jet needles (thinner) and shorter springs than the No. 2 and No. 4 carburetors (SABRE only). Do not interchange these parts.

TOOLS

Special

Hand vacuum pump/gauge
Valve guide driver, 7 mm

ST-AH-260-MC7 or A973X-041-XXXXX (U.S.A. only)
07942-8230000 (U.S.A. only)

Common

Float gauge

07401-0010001

SPECIFICATIONS

	V45 SABRE	V45 MAGNA
Venturi dia.	Primary 11.7 mm (0.46 in) Secondary 29.2 mm (1.15 in)	Primary 11.1 mm (0.44 in) Secondary 29.2 mm (1.15 in)
Identification No.	VD50A	VD51A
Float level	8.3 mm (0.33 in)	7.2 mm (0.28 in)
Main jet	# 132	# 128
Idle speed	1,000 ± 100 rpm	
Throttle grip free play	2-6 mm (0.08-0.24 in)	
Pilot screw initial opening	See page 4-18	



TROUBLESHOOTING

Engine cranks but won't start

1. No fuel in tank
2. No fuel to carburetor
3. Engine flooded with fuel
4. No spark at plug (ignition system faulty)
5. Air cleaner clogged
6. Intake air leak
7. Improper choke operation
8. Improper throttle operation

Hard starting or stalling after starting

1. Improper choke operation
2. Ignition malfunction
3. Carburetor faulty
4. Fuel contaminated
5. Intake air leak
6. Idle speed incorrect

Rough idle

1. Ignition system faulty
2. Idle speed incorrect
3. Incorrect carburetor synchronization
4. Carburetor faulty
5. Fuel contaminated

Misfiring during acceleration

1. Ignition system faulty

Backfiring

1. Ignition system faulty
2. Carburetor faulty

Poor performance (driveability) and poor fuel economy

1. Fuel system clogged
2. Ignition system faulty

Lean mixture

1. Clogged fuel jets
2. Piston stuck closed
3. Faulty float valve
4. Float level low
5. Fuel cap vent blocked
6. Fuel strainer screen clogged
7. Restricted fuel line
8. Air vent tube clogged
9. Intake air leak
10. Restricted or faulty fuel pump (V45 MAGNA)

Rich mixture

1. Clogged air jets
2. Faulty float valve
3. Float level too high
4. Choke stuck clogged
5. Dirty air cleaner



CARBURETOR REMOVAL (V45 SABRE)

Turn the fuel valve OFF.

Remove the seat, frame side covers and fuel tank.

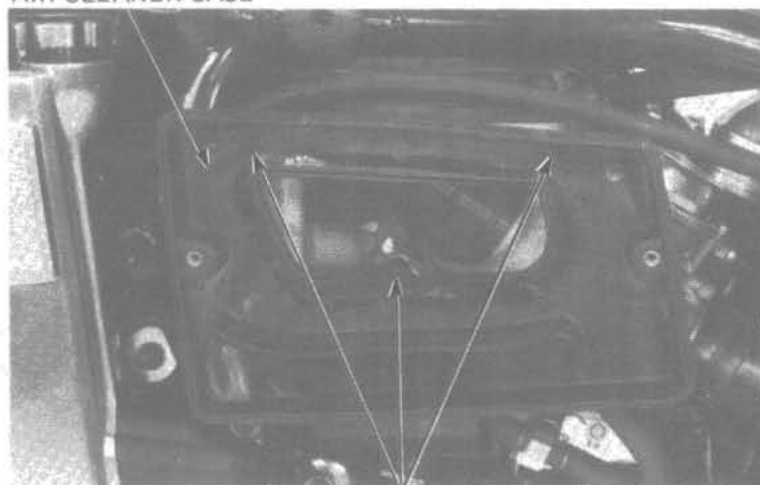
Remove the right and left air cleaner covers and the air cleaner elements.

AIR CLEANER CASE COVER



Remove both air cleaner case screws and air cleaner cases.

AIR CLEANER CASE



SCREWS

IGNITION COIL
MOUNTING BOLT

Remove the right ignition coil mounting bolts.
Remove the screw attaching the water pipe to the air chamber.
Remove the air chamber cover screws and cover.
Remove the radiator side mounting bolts.

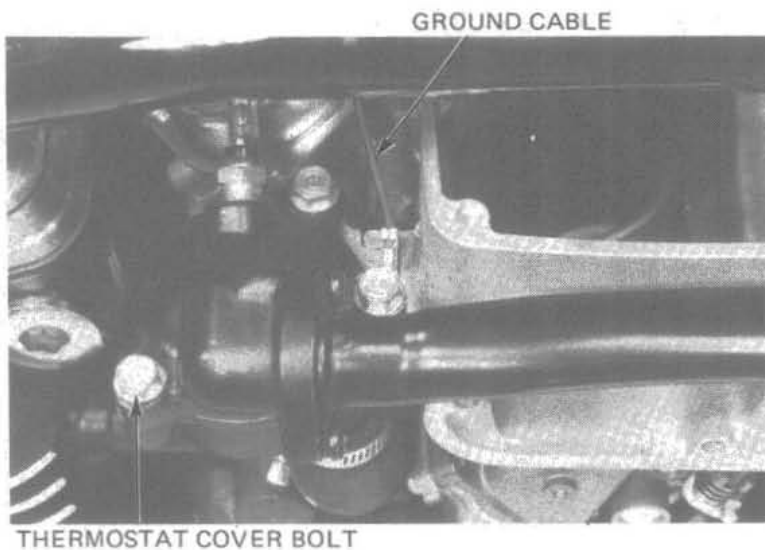


AIR CHAMBER COVER

WATER PIPE
ATTACHING SCREW



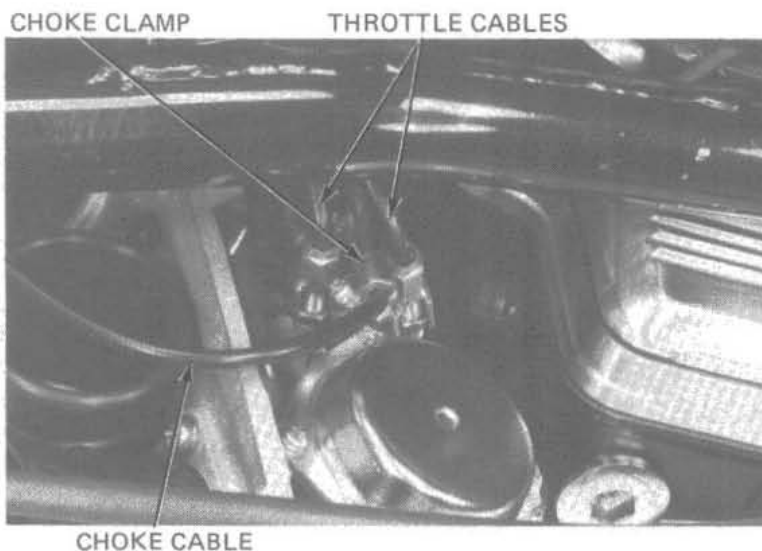
Remove the No. 1 carburetor intake pipe bands.
Remove the thermostat cover bolt and disconnect the ground cable.
Remove the temperature sensor wire.
Disconnect the air breather tube.



Loosen all the carburetor bands. Remove the carburetors from the intake pipes. Also, remove the No. 1 intake pipe from the cylinder head.
Disconnect the spark plug wire clamps on the heat shield.
Remove the carburetor assembly from the left side.



Remove the carburetor choke clamp and disconnect the choke and throttle cables.





CARBURETOR REMOVAL (V45 MAGNA)

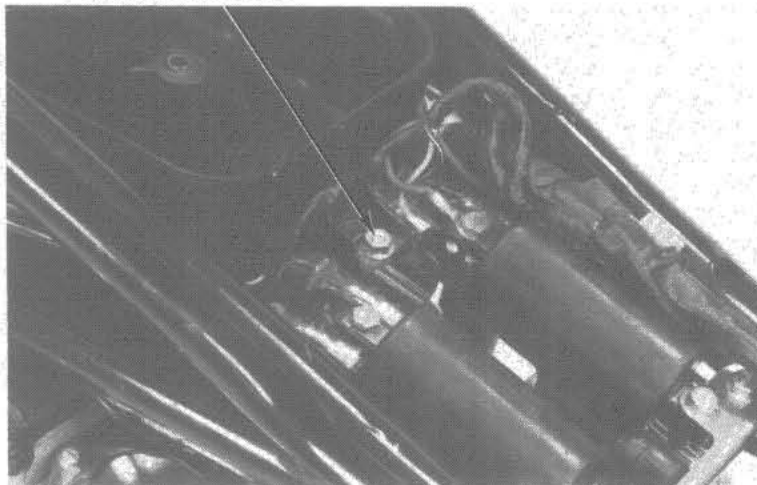
Check that the fuel tank is at least one-half empty. Otherwise, drain the fuel tank until it is one-half empty before raising the tank.

Remove the seat and front fuel tank mounting bolts. Lift the fuel tank up and put the support arm down.

Remove the air cleaner cover and element.

Remove the air cleaner case mounting bolt and case.

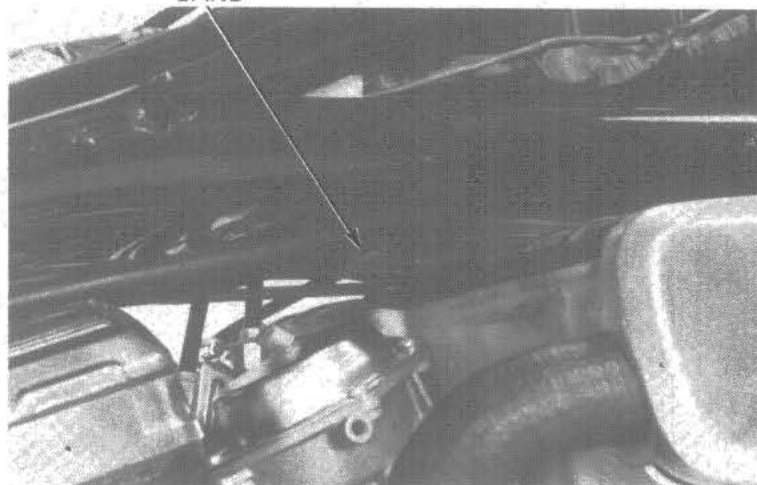
**AIR CLEANER CASE
MOUNTING BOLT**



Loosen the air cleaner connecting bands and remove the air cleaner case.

Disconnect the spark plug lead clamps on the heat shield.

BAND

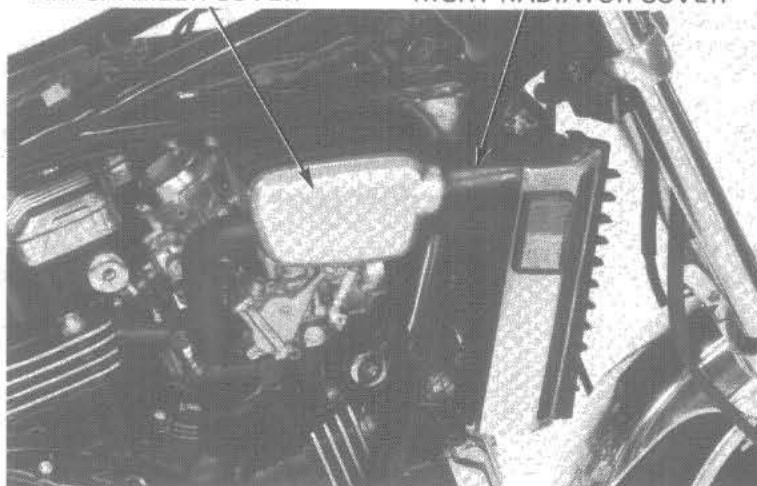


Remove the radiator right upper hose cover. Remove the right and left air chamber covers. Drain and remove the radiator (page 6-8).

Loosen the clamps on the Nos. 1, 3 and Nos. 2, 4 cylinder coolant crossover pipes. Remove the crossover pipes.

AIR CHAMBER COVER

RIGHT RADIATOR COVER



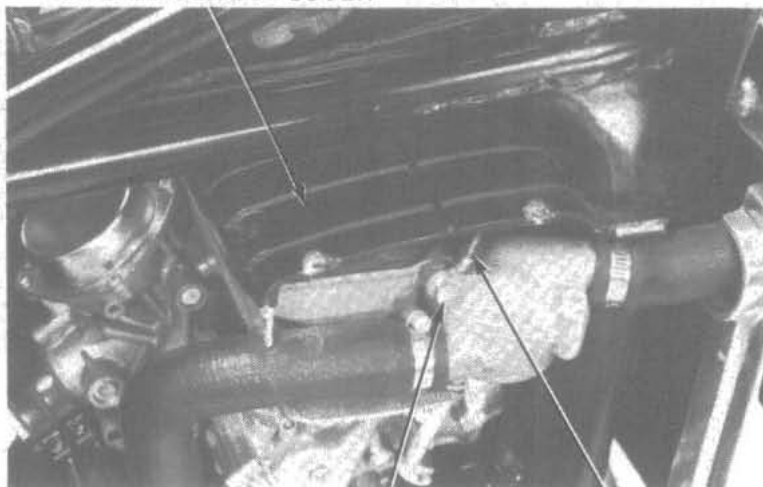


Remove the air chamber top cover.
Remove the bolt attaching the thermostat housing to the air chamber and the ground cable. Remove the thermostat housing and coolant hoses.
Remove the temperature sensor wire.
Disconnect the breather tube and loosen the carburetor bands.
Remove all the No. 1 and No. 3 carburetor intake pipe bands.
Loosen the No. 2 and No. 4 carburetor intake pipe bands, on the carburetor side only.
Remove the carburetor assembly from the intake pipes.
Remove the No. 1 and No. 3 intake pipes from the cylinder head.
Remove the carburetor choke clamp and disconnect the choke and throttle cables.
Remove the carburetor assembly from the right side.

NOTE:

It may be necessary to loosen the engine mount bolts and move the engine on its rubber mounts to provide clearance for carburetor removal.

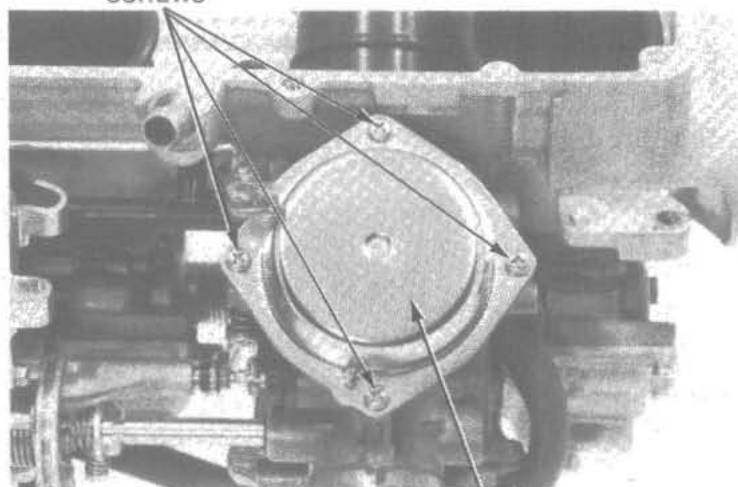
AIR CHAMBER TOP COVER



BOLT

GROUND CABLE

SCREWS



COVER

VACUUM CHAMBER

REMOVAL

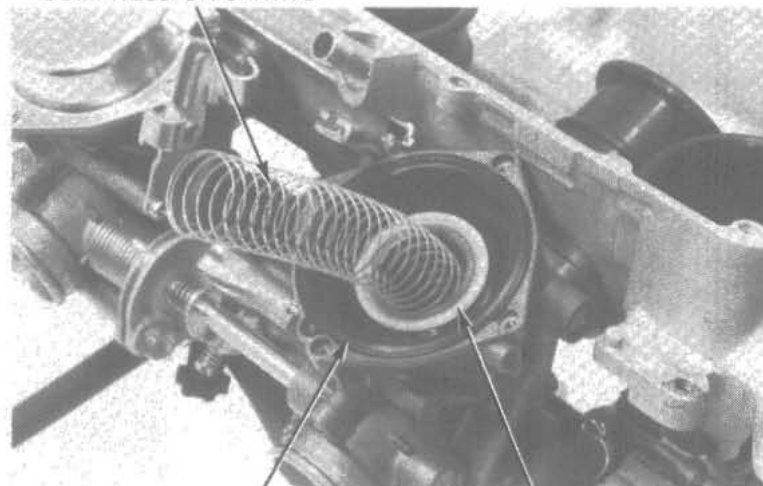
Remove the four vacuum chamber cover screws and cover.

CAUTION

Do not interchange vacuum chamber covers, springs pistons or jet needles between carburetors.

Remove the compression spring, diaphragm and vacuum piston.
Inspect the vacuum piston for wear, nicks, scratches or other damage.
Make sure the piston moves up and down freely in the chamber.
Note that the No. 1 and No. 3 carburetors use thinner jet needles and shorter springs than the No. 2 and No. 4 carburetors (SABRE only).

COMPRESSION SPRING

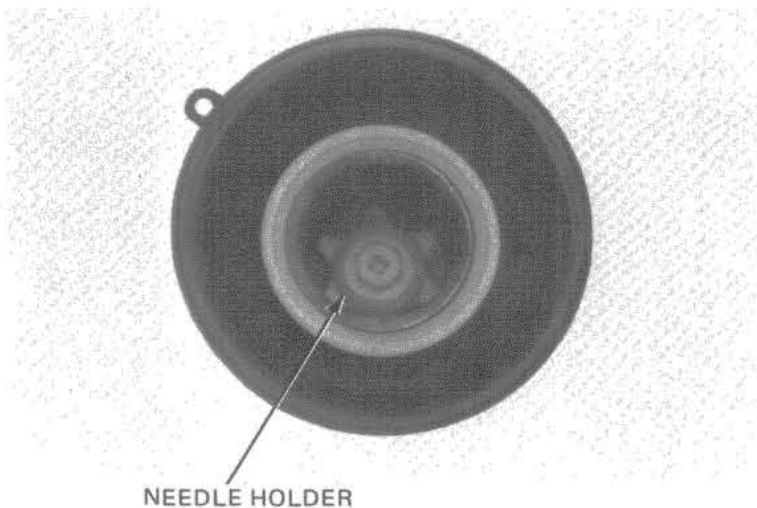


DIAPHRAGM

VACUUM PISTON

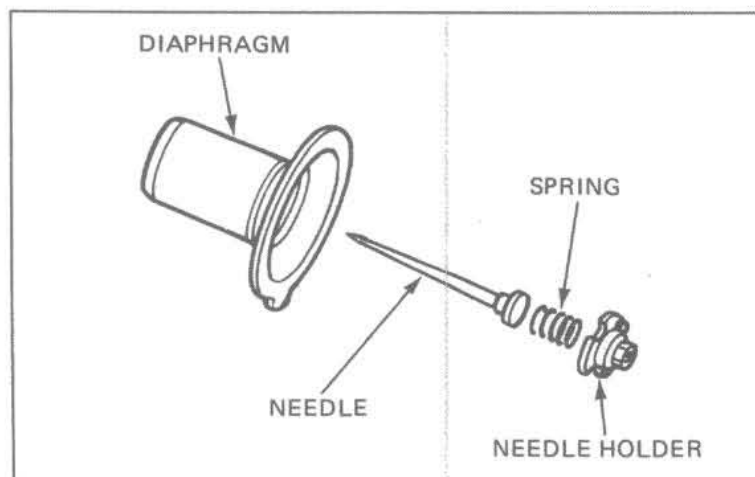


Push the needle holder in and turn it 60 degrees with an 8 mm socket. Then remove the needle holder, spring and needle from the piston.



Inspect the needle for excessive wear at the tip and for bending, or other damage.

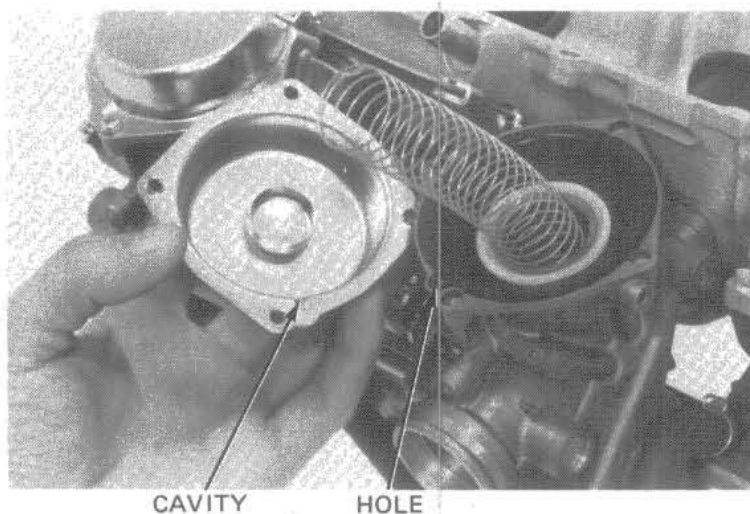
Check the diaphragm for deterioration and tears.



INSTALLATION

Installation is essentially the reverse of removal. Install the chamber cover so that its cavity aligns with the hole in the diaphragm.

Be sure the thinner jet needles and shorter springs are installed in the No. 1 and No. 3 carburetors.

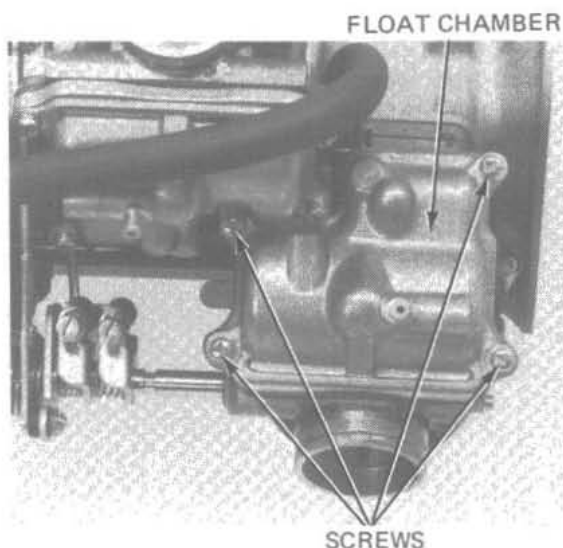




FLOAT CHAMBER

REMOVAL

Remove the four float chamber screws and the float chamber.



FLOAT LEVEL

Measure the float level with the carburetor inclined 15° – 45° from vertical and the float tang just contacting the float valve.

Specification:

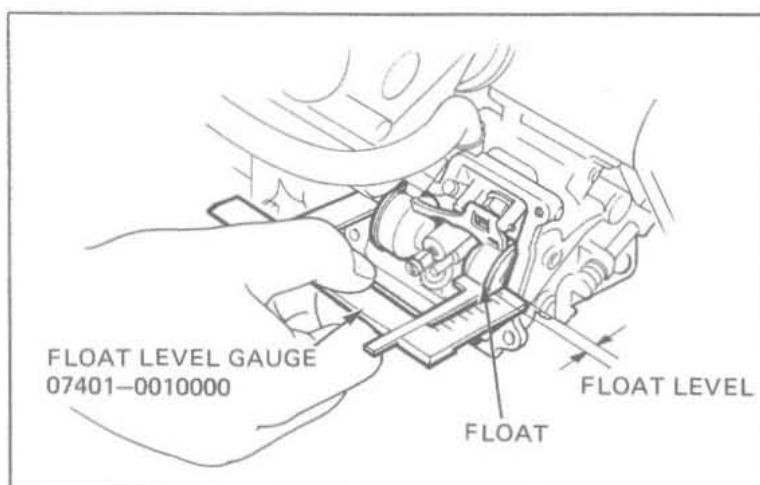
V45 Magna—7.2 mm (0.28 in)

V45 Sabre—8.3 mm (0.33 in)

NOTE:

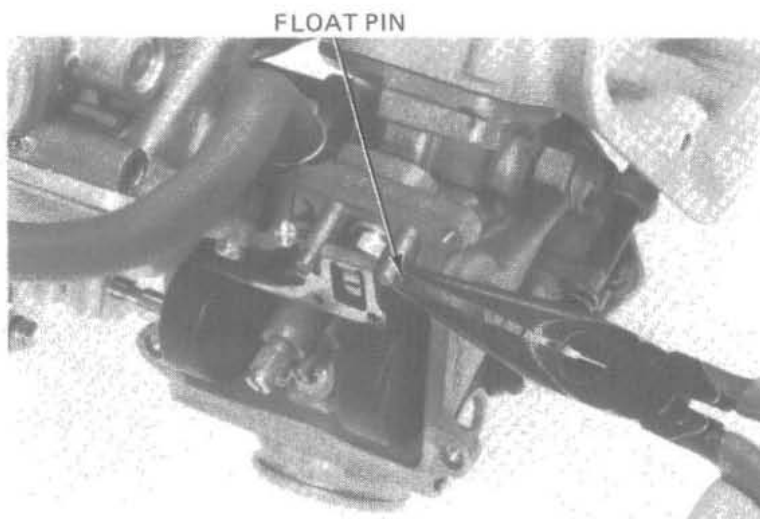
Measure the float nearest the float valve.

Adjust the float level by carefully bending the float tang.



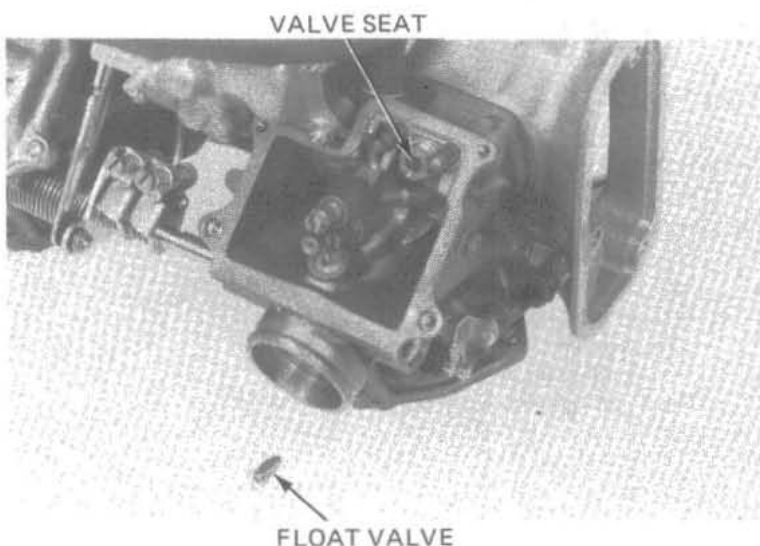
FLOAT AND JETS

Remove the float pin, float and float valve.

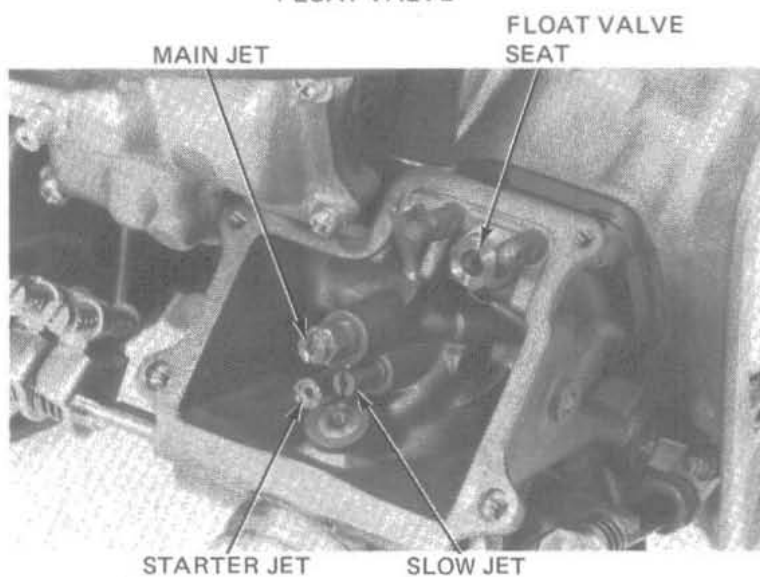




Inspect the float valve for grooves and nicks.
Inspect the operation of the float valve.



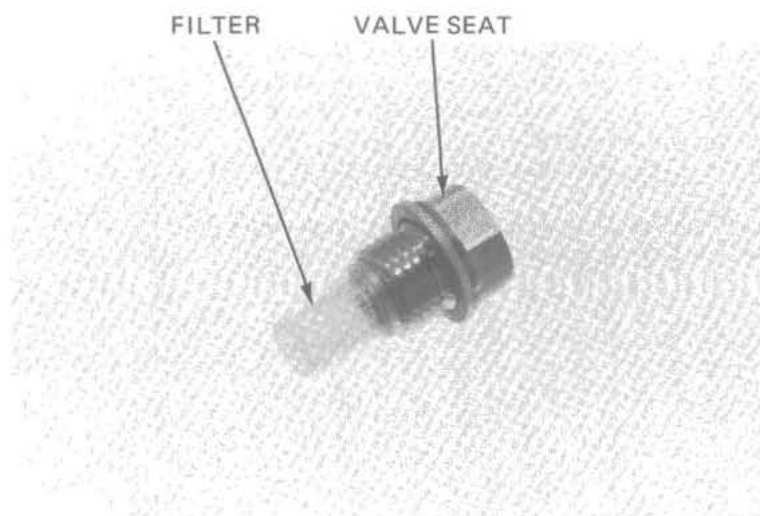
Remove the starter jet, main jet and slow jet.
Remove the float valve seat and filter.



Inspect the float valve seat and filter for grooves,
nicks or deposits.

ASSEMBLY

Assemble the float chamber components in the
reverse order of disassembly.





PILOT SCREW

REMOVAL

NOTE:

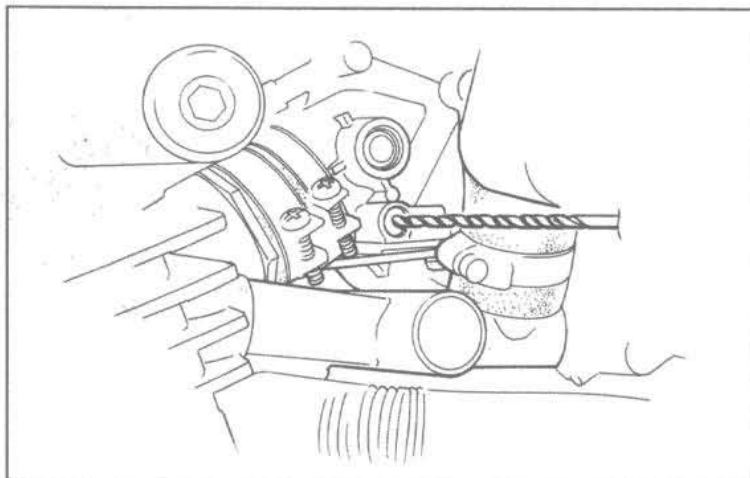
- The pilot screws are factory pre-set and should not be removed unless the carburetors are overhauled.
- The pilot screw plugs are factory installed to prevent pilot screw misadjustment. Do not remove the plugs unless the pilot screws are being removed.
- Cover all opening with tape to keep metal particles out when the plugs are drilled.

Center punch the pilot screw plug to center the drill point.

Drill through the plug with a 4 mm (5/32 in) drill bit, being careful not to drill into the pilot screw.

CAUTION:

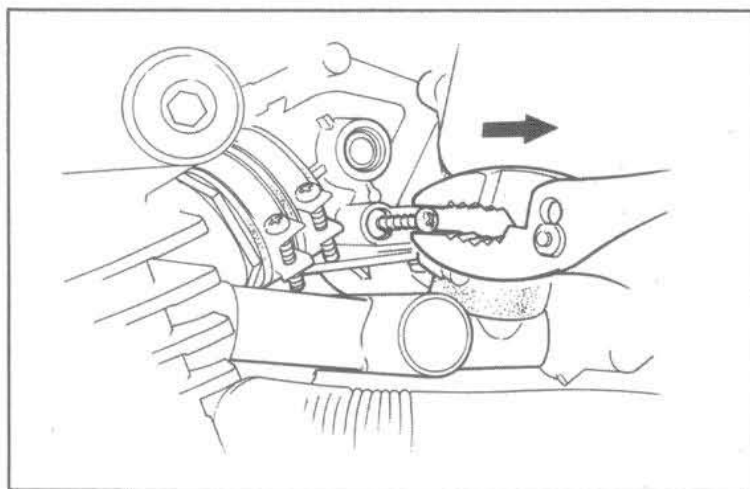
Be careful not to drill into the pilot screw. All pilot screws must be replaced even if only one requires it for proper pilot screw adjustment (page 4-18).



Force a self-tapping 4 mm screw (H/C 069399, P/N 93903-35410) into the drilled plug and continue turning the screwdriver until the plug rotates with the screw.

Pull on the screw head with pliers to remove the plug.

Use compressed air to clean the pilot screw area and remove metal shavings.





Turn each pilot screw in and carefully count the number of turns before it seats lightly. Make a note of this to use as a reference when reinstalling the pilot screws.

CAUTION

Damage to the pilot screw seat will occur if the pilot screw is tightened against the seat.

Remove the pilot screws and inspect them. Replace them if they are worn or damaged.

INSTALLATION

Install the pilot screws and return them to their original position as noted during removal. Perform pilot screw adjustment if new pilot screws are installed (page 4-18).

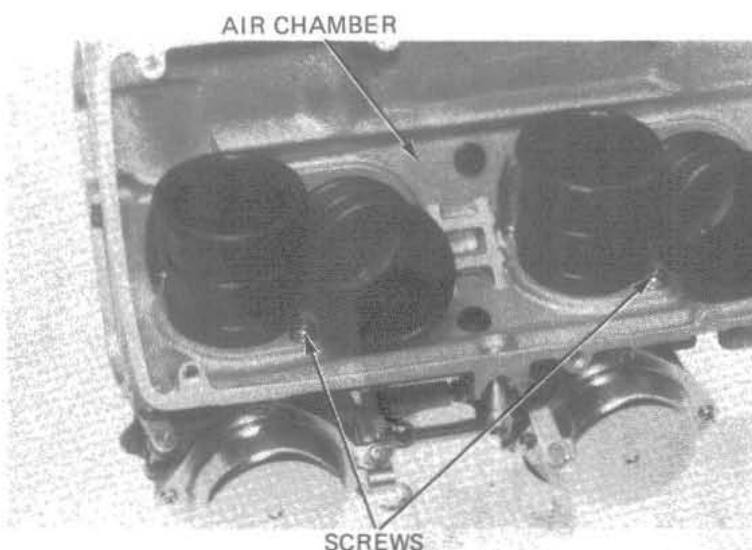
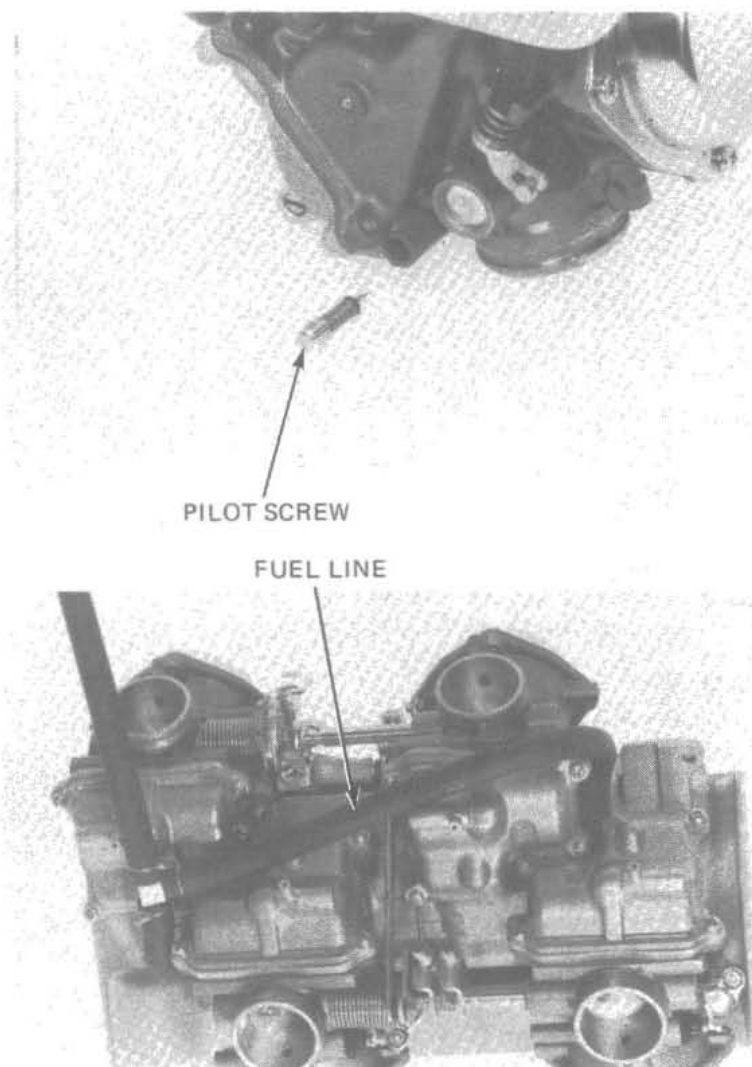
NOTE

- Do not install new plugs on new pilot screw holes until after adjustment has been made.
- If you replace the pilot screw in one carburetor, you must replace the pilot screws in the other carburetors for proper pilot screw adjustment.

CARBURETOR SEPARATION

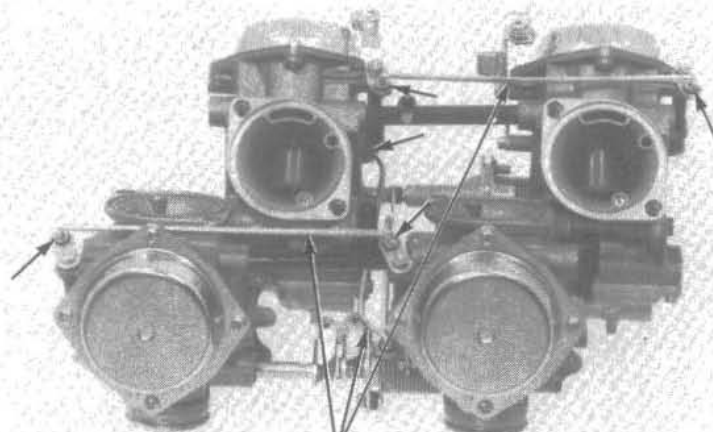
Remove the fuel lines from the carburetors.

Remove the screws attaching the air chamber to the carburetors and separate the chamber and carburetors.





Remove the choke rod cotter pins and remove the choke rods.

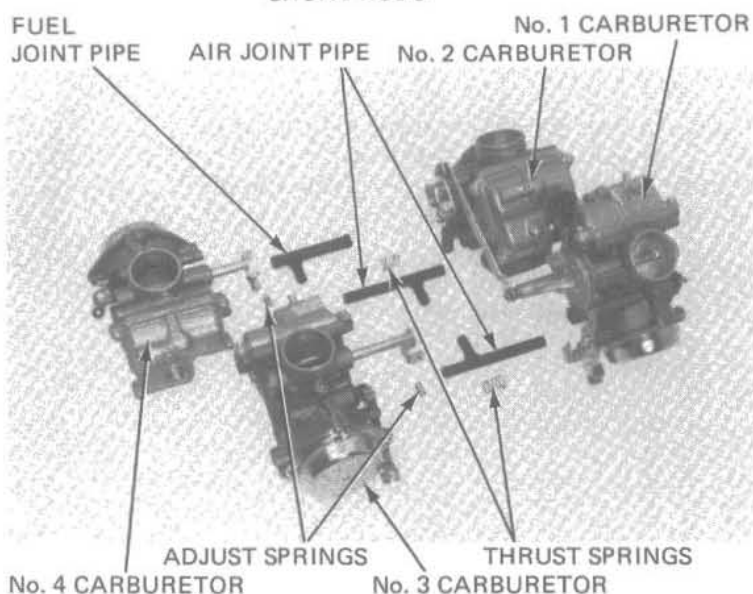


CHOKE RODS

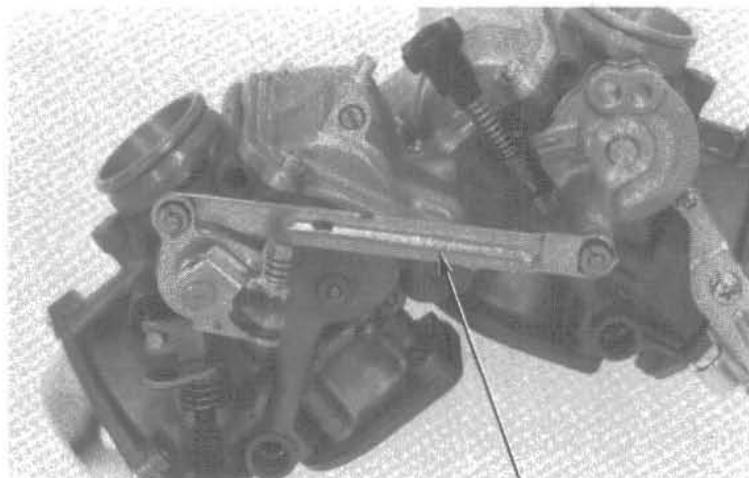
Carefully separate the No. 3 carburetor from the assembly. Then separate the No. 4 carburetor.

CAUTION

Separate the carburetors horizontally to prevent damage to the joint pipes.



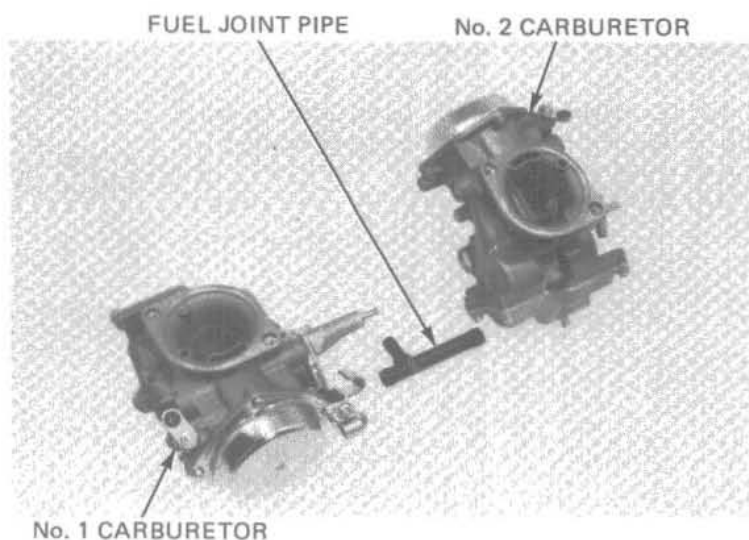
Disconnect the throttle link from the No. 1 and 2 carburetors by removing the cotter pins.



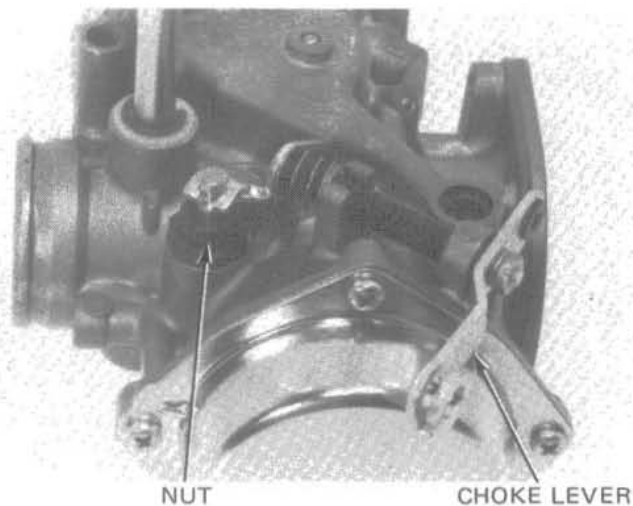
THROTTLE LINK



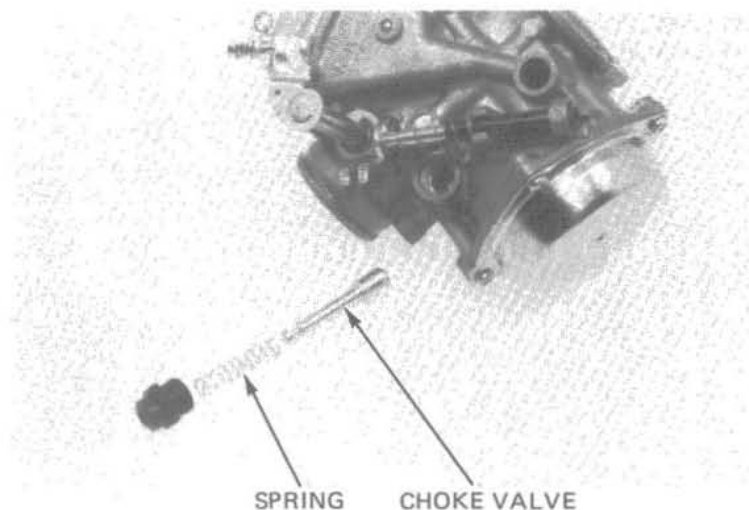
Carefully separate the No. 1 and No. 2 carburetors.



Remove the choke lever from the carburetor.
Remove the choke valve nut, spring and valve.



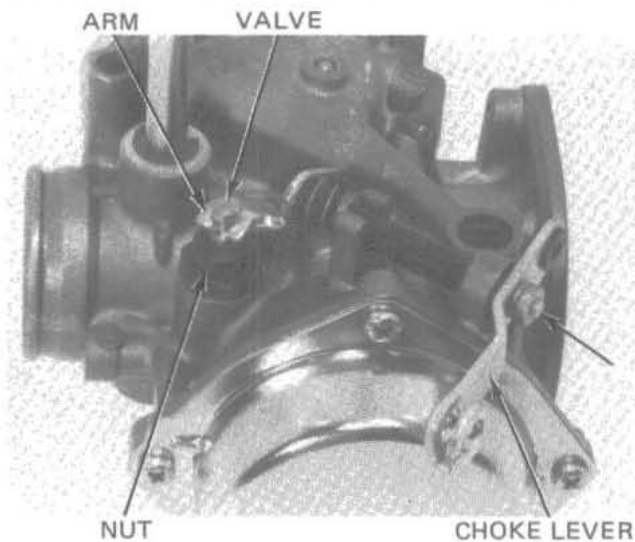
Check the choke valve and spring for nicks, grooves,
or other damage.



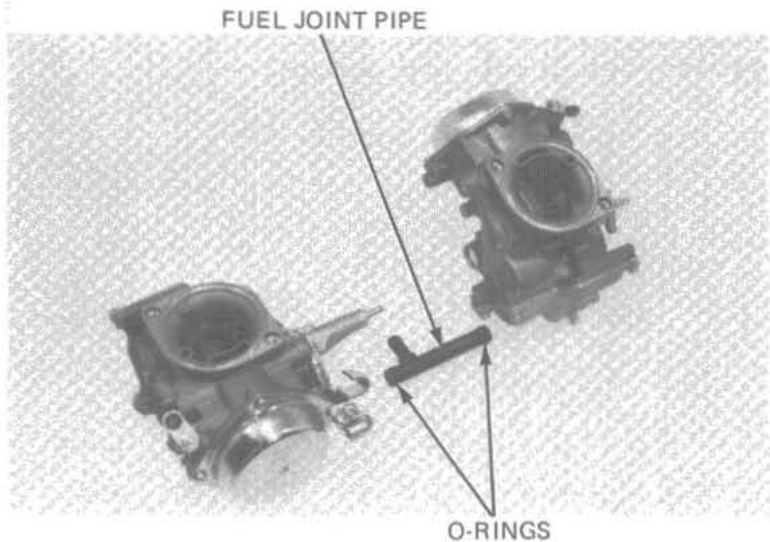


CARBURETOR ASSEMBLY

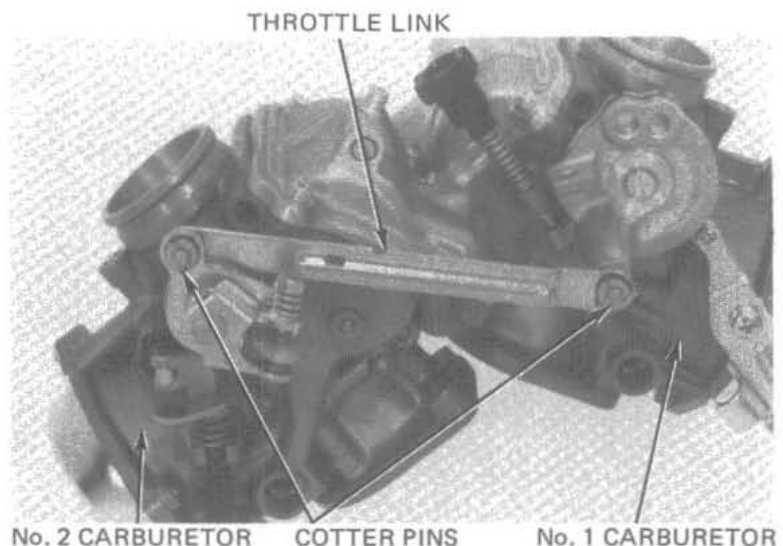
Install the choke valve, valve spring and nut and tighten the nut while hooking the choke arm to the groove in the valve.
 Install the choke lever.



Coat the new O-rings with oil and install them on the fuel joint pipe for No. 1 and No. 2 carburetors.
 Install the fuel joint pipe to the No. 1 and No. 2 carburetors.



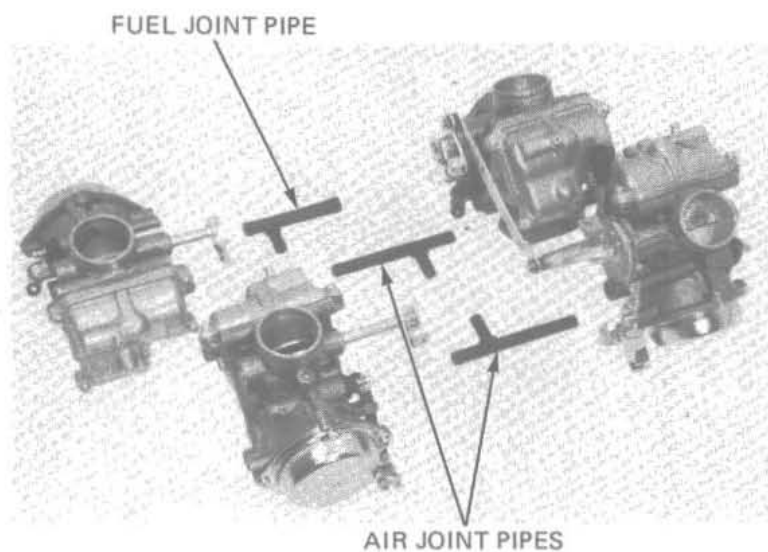
Reconnect the throttle linkage between the No. 1 and No. 2 carburetors, using new cotter pins.





Coat new O-rings with oil and install them on the fuel and air joint pipes.

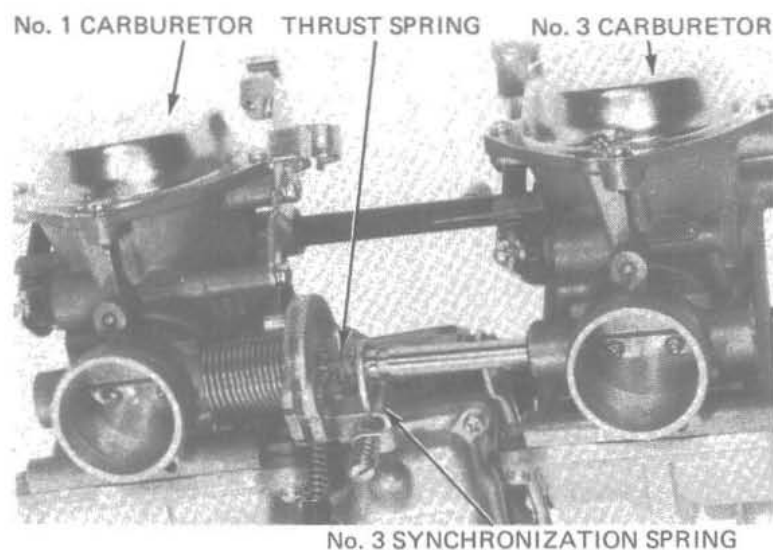
Put the No. 3 and No. 4 carburetors together with the joint pipes.



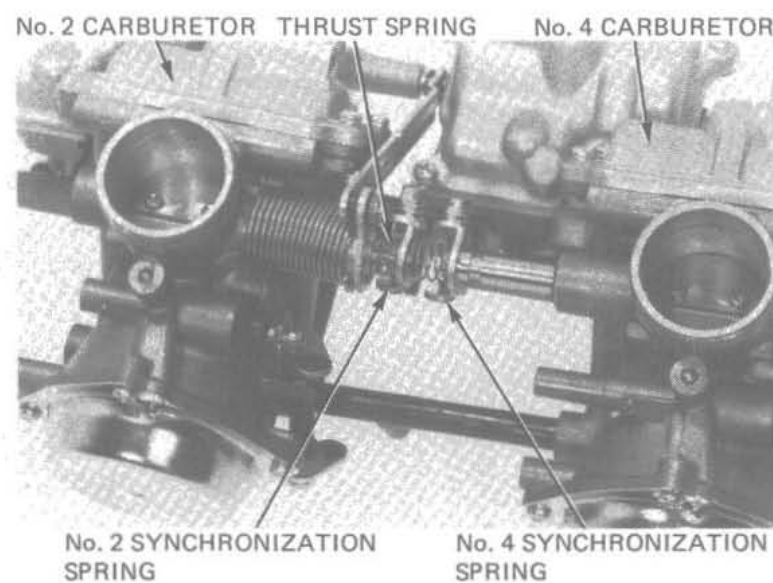
Loosen the synchronization adjusting screws until there is no tension.

Install the synchronization springs.

Install the thrust springs between the throttle valve shafts.

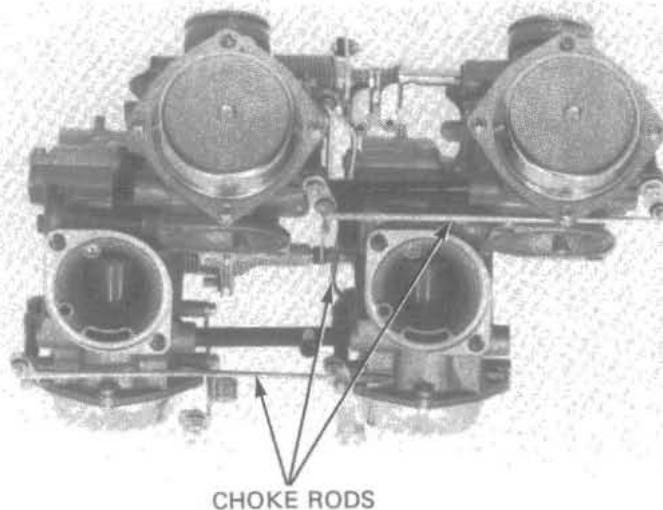


Make sure the fuel joint and air joint pipes are securely installed.

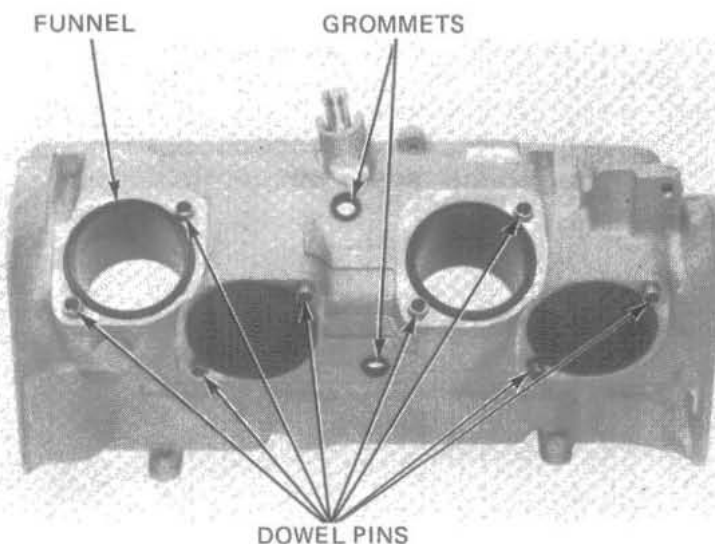




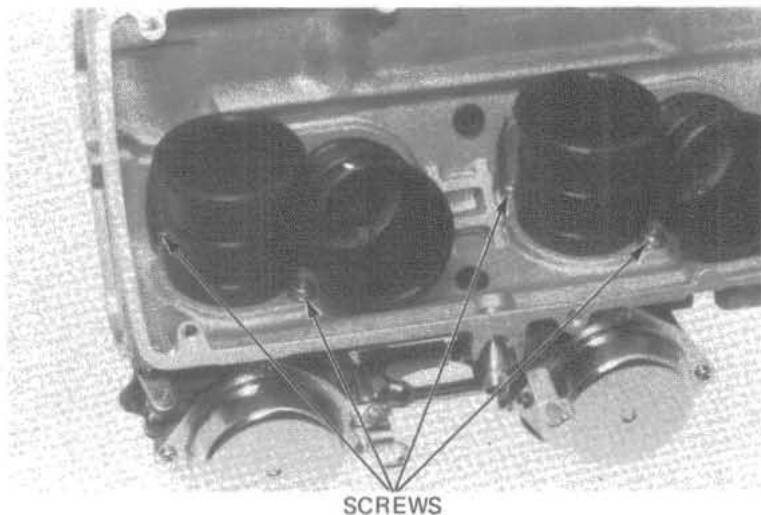
Install the choke rods, using new cotter pins.



Make sure the air chamber funnels, grommets and dowel pins are in place.



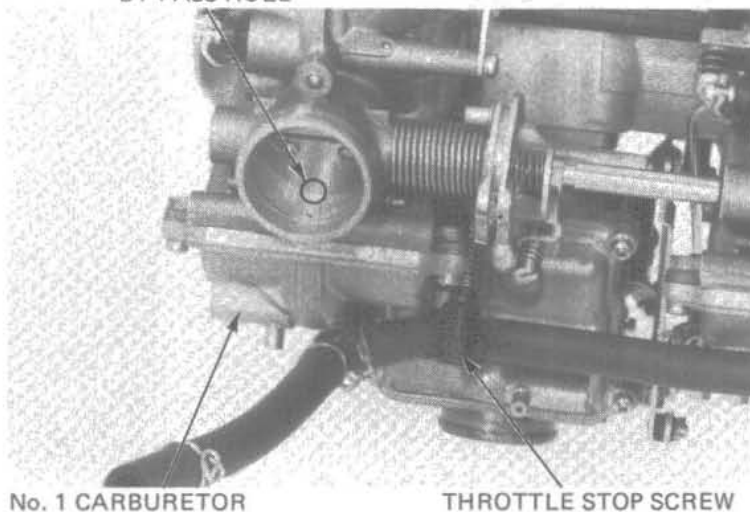
Place the air chamber over the carburetors aligning the dowel pins with the carburetor holes.
Attach the air chamber to the carburetors with the eight screws.
Connect the fuel lines to the fuel joint pipes.





Turn the throttle stop screw to align the No. 1 throttle valve with the edge of the by-pass hole.

BY-PASS HOLE



No. 1 CARBURETOR

THROTTLE STOP SCREW

Align each throttle valve with the by-pass hole edge by turning the synchronization adjusting screws.

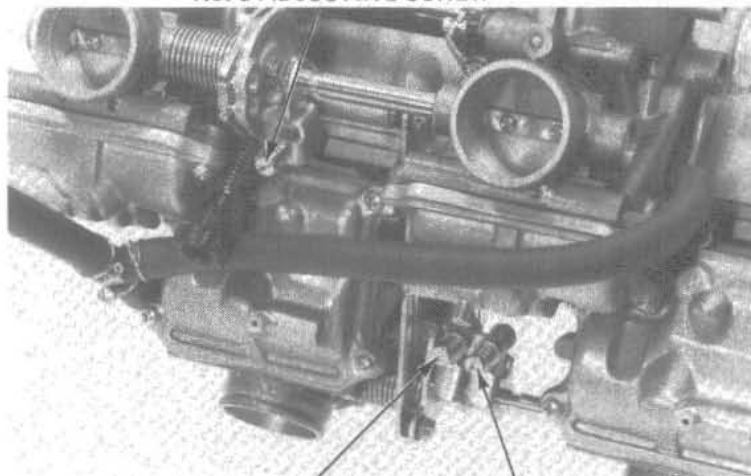
Inspect throttle operation as described below:

- Open the throttle slightly by pressing the throttle linkage. Then release the throttle.
- Make sure that it returns smoothly.
- Make sure that there is no drag when opening and closing the throttle.

Make sure that choke valve operation is smooth by moving the choke linkage.

Close the choke valve by turning the choke linkage. Release the choke linkage and make sure that it returns smoothly.

No. 3 ADJUSTING SCREW



No. 2 ADJUSTING SCREW

No. 4 ADJUSTING SCREW

CARBURETOR INSTALLATION

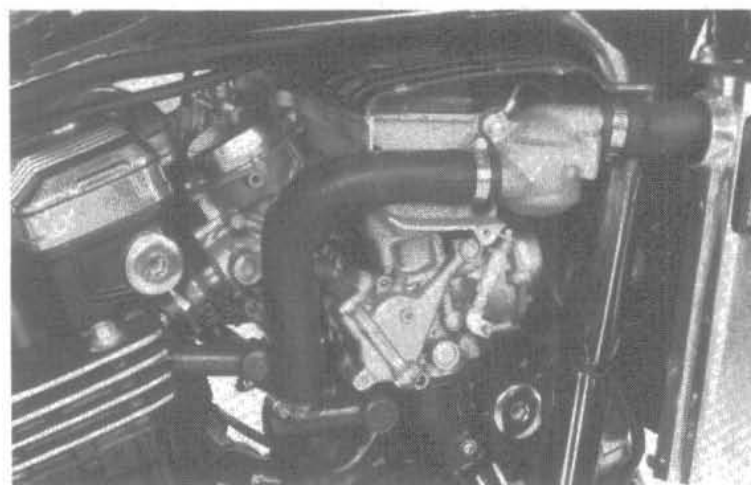
Installation is essentially the reverse of removal.

NOTE

Route the throttle and choke cables properly (page 1-9 to 1-10).

Perform the following inspections and adjustments.

- Throttle operation (page 3-6).
- Carburetor choke (page 3-7)
- Carburetor idle speed (page 3-13)
- Carburetor synchronization (page 3-12).





PILOT SCREW ADJUSTMENT

IDLE DROP PROCEDURE (U.S.A. ONLY)

NOTE

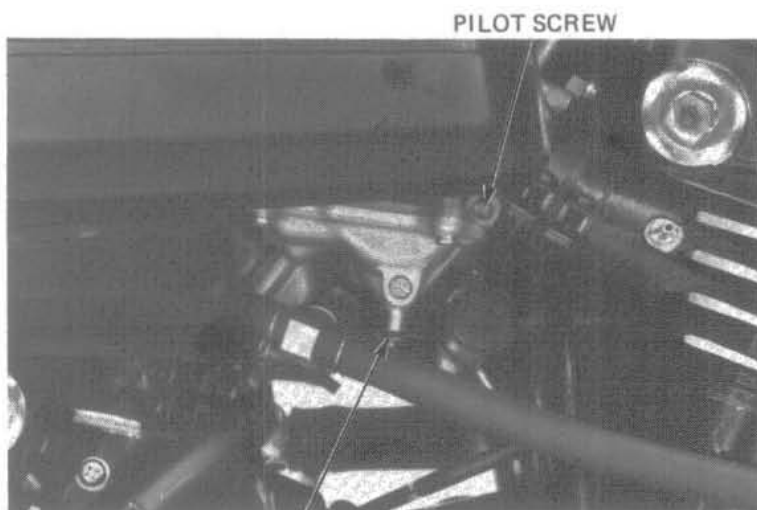
- The pilot screws are factory pre-set and no adjustment is necessary unless the pilot screws are replaced (page 4-10).
- Use a tachometer with graduations of 50 rpm or smaller that will accurately indicate a 50 rpm change.

1. Turn each pilot screw clockwise until it seats lightly and back it out to the specification given. This is an initial setting prior to the final pilot screw adjustment.

INITIAL OPENING: 2-3/4 turns out

CAUTION:

Damage to the pilot screw seat will occur if the pilot screw is tightened against the seat.

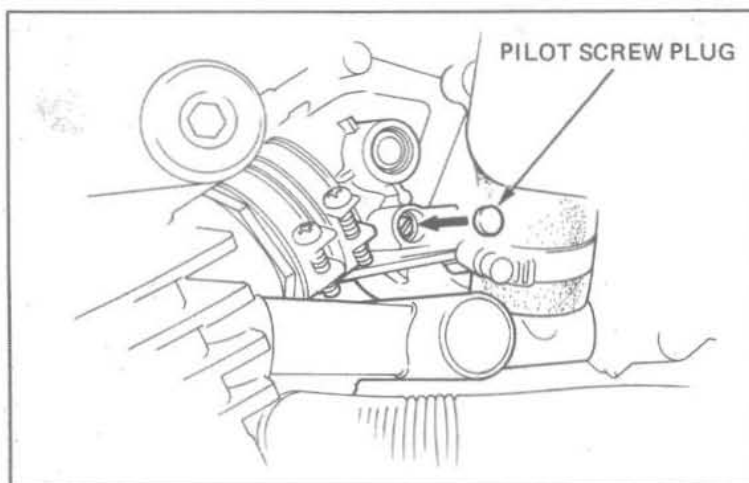


THROTTLE STOP SCREW

2. Warm up the engine to operating temperature. Stop and go driving for 10 minutes is sufficient.
3. Attach a tachometer according to the manufacturer's instructions.
4. Adjust the idle speed with the throttle stop screw.

IDLE SPEED: 1,000 ± 100 rpm

5. Turn each pilot screw 1/2 turn out from the initial setting.
6. If the engine speed increases by 50 rpm or more, turn each pilot screw out by a continual 1/2 turn until engine speed drops by 50 rpm or less.
7. Adjust the idle speed with the throttle stop screw.
8. Turn the No. 1 carburetor pilot screw in until the engine speed drops 50 rpm.
9. Turn the No. 1 carburetor pilot screw 1 turn out from the position obtained in step 8.
10. Adjust the idle speed with the throttle stop screw.
11. Perform steps 8, 9 and 10 for the No. 2, 3 and 4 carburetor pilot screws.
12. Drive new pilot screw plugs into the pilot screw bores with a 7 mm valve guide driver (P/N 07942-8230000). When fully seated the plug surfaces will be recessed 1 mm into the pilot screw bore.





FUEL TANK

WARNING

*Do not allow flames or sparks near gasoline.
Wipe up spilled gasoline at once.*

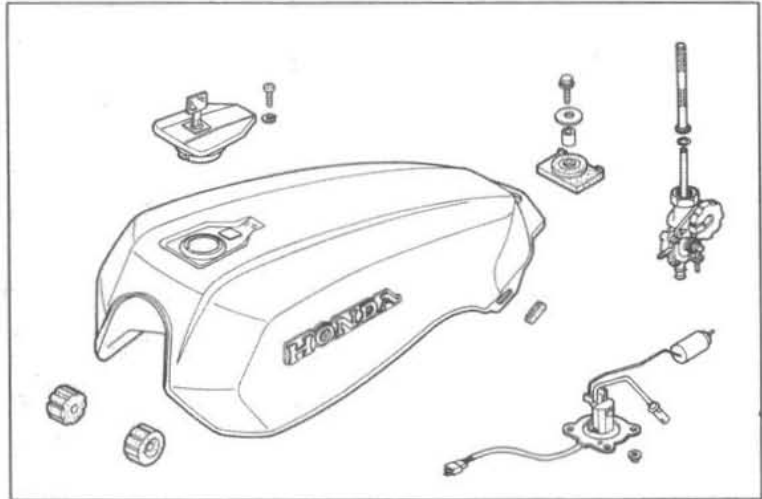
Check the vent hole of the filler cap for blockage. Check that fuel is flowing out of the fuel valve freely. If fuel flow is restricted, clean the fuel strainer.

NOTE

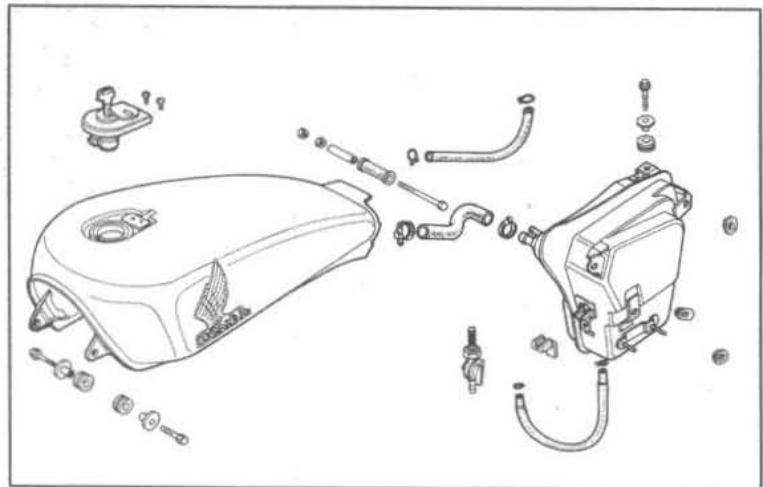
Do not overtighten the fuel valve lock nut (V45 SABRE).

Make sure that there are no fuel leaks.

V45 SABRE



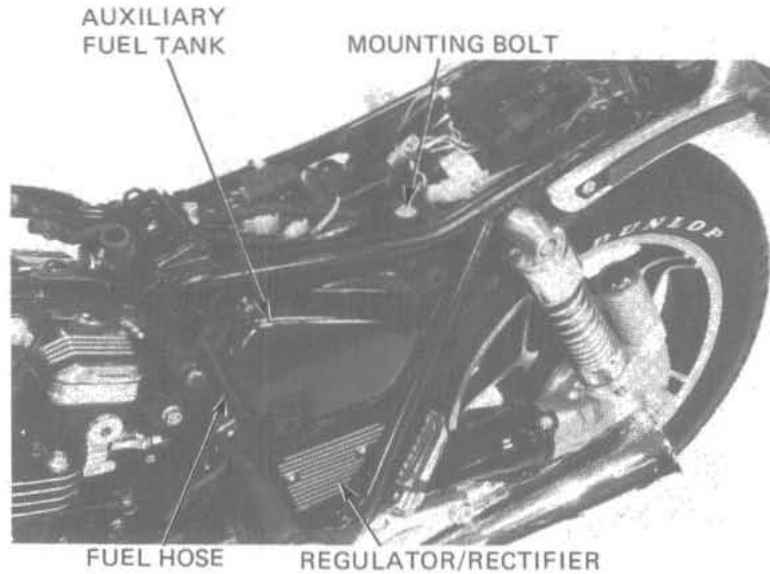
V45 MAGNA





AUXILIARY FUEL TANK (V45 MAGNA)

Remove the seat, main fuel tank and side covers.
 Remove the regulator/rectifier.
 Disconnect the battery negative cable, then positive cable and remove the battery.
 Remove the rear wheel (page 17-3).
 Remove rear fenders A and B.
 Detach the auxiliary fuel tank hose from the fuel pump.
 Remove the auxiliary fuel tank mounting bolt and tank.
 Install the auxiliary fuel tank in the reverse order of removal.

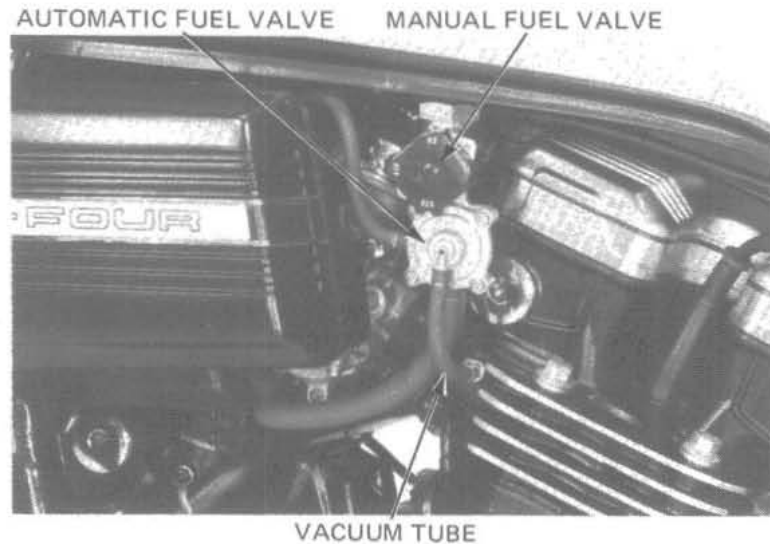


AUTOMATIC FUEL VALVE (V45 SABRE)

WARNING

*Do not allow flames or sparks near gasoline.
 Wipe up spilled gasoline at once.*

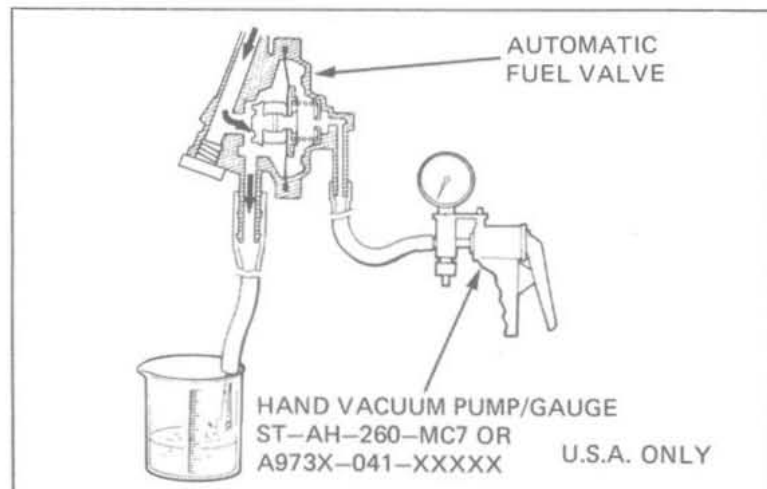
Turn the manual valve ON.
 Stop the engine and place a container under the fuel line.
 Remove the fuel tube from the carburetor.
 Inspect the fuel tube for fuel flowing out.
 Drain the remaining fuel in the fuel valve and fuel tube (approx. 5–10 cc).
 If the fuel flow stops, the automatic fuel valve is operating normally. If the fuel flow does not stop, clean the vacuum tube.



Disconnect the vacuum tube from the intake port and apply vacuum with a vacuum pump.
 The valve is operating normally if fuel flows out of the tube when vacuum is applied and fuel stops flowing out when the vacuum pump is disconnected.

If the valve does not operate normally:

- Inspect for clogging and clean the valve.
- Blow air through the valve from the inlet side to set the flat diaphragm in its original position.





AIR CLEANER

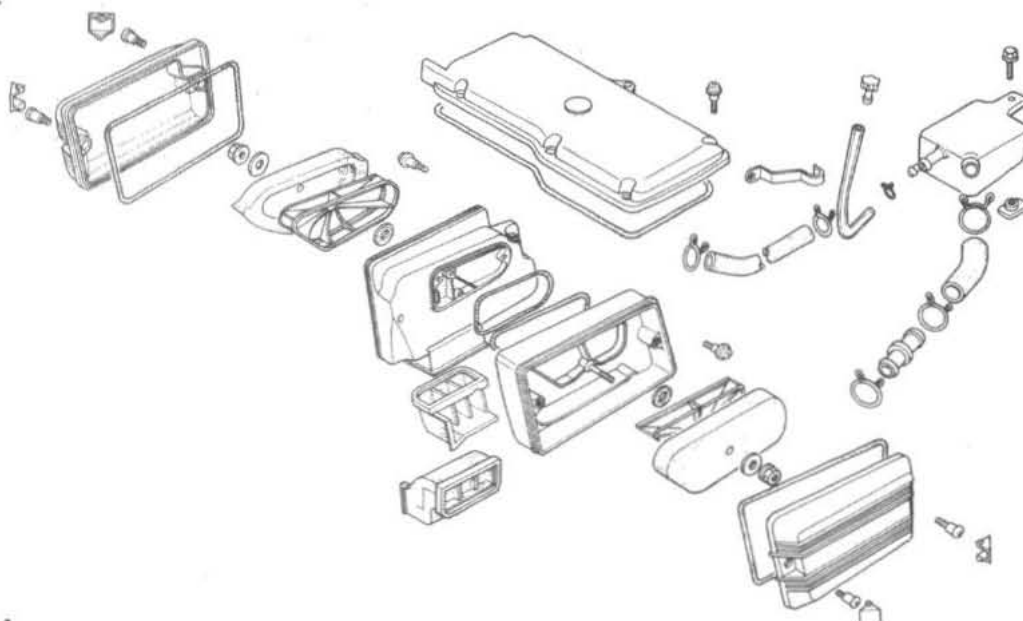
CASE/CHAMBER

Check the air cleaner case for deterioration.

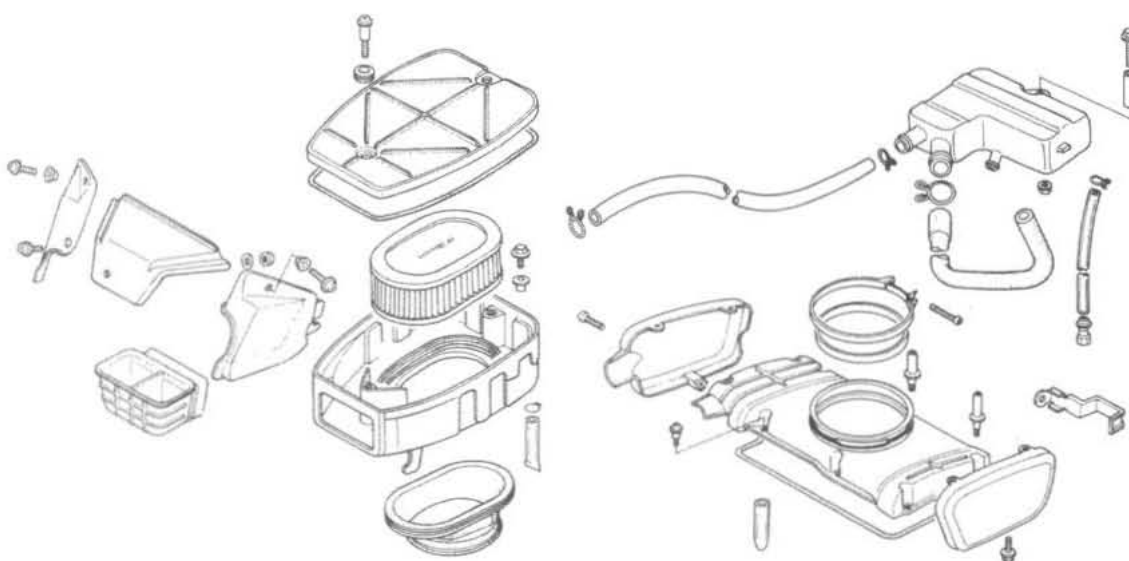
CRANKCASE VENTILATION SYSTEM

Check that the breather tube is not restricted.

V45 SABRE



V45 MAGNA





FUEL PUMP (V45 MAGNA)

Remove the seat, fuel tank (page 4-19) and side covers.

Disconnect the battery negative cable, then positive cable and remove the battery.

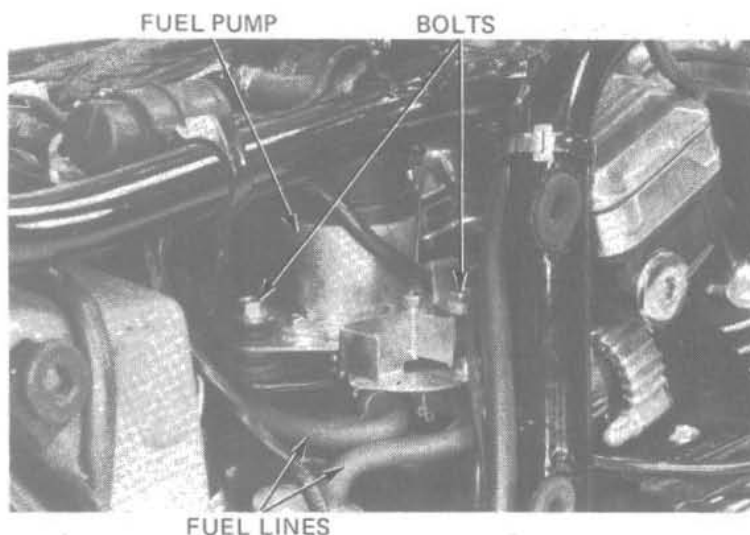
Remove the battery tray and starter relay switch. Turn the fuel valve off.

Detach the fuel inlet and outlet lines from the fuel pump.

Disconnect the fuel pump wire connectors.

Remove the fuel pump mounting bolts and fuel pump.

Install the fuel pump in the reverse order of removal.



HIGH ALTITUDE ADJUSTMENT (USA only)

When the vehicle is to be operated continuously above 2,000 m (6,500 feet) the carburetor must be readjusted as follows to improve driveability and decrease exhaust emissions.

Warm up the engine to operating temperature. Stop and go driving for 10 minutes is sufficient.

Remove each pilot screw plug (page 4-10).

Turn each pilot screw clockwise 1 turn.

Adjust the idle speed to $1,000 \pm 100$ rpm with the throttle stop screw.

Drive new pilot screw plugs into the pilot screw bores (page 4-18).

NOTE:

This adjustment must be made at high altitude to ensure proper high altitude operation.

Attach a Vehicle Emission Control Information Update label onto the frame as shown. Refer to Service Bulletin #SL132 for information on obtaining the label.

NOTE:

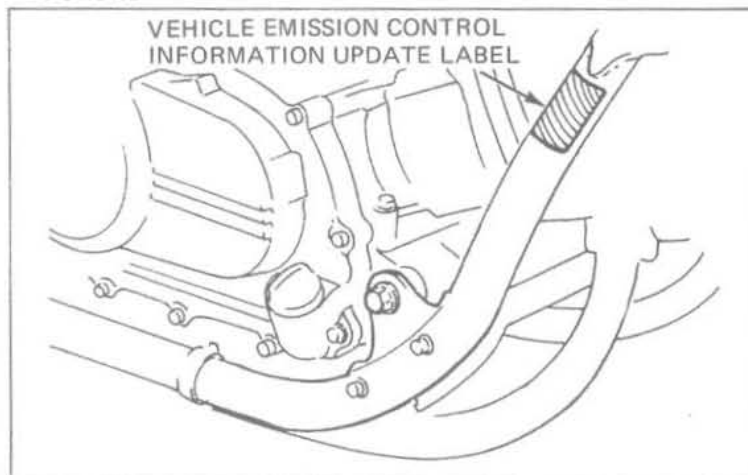
Do not attach the label to any part that can be easily removed from the vehicle.

WARNING

Operation at an altitude lower than 1,500 m (5,000 feet) with the carburetors adjusted for high altitudes may cause the engine to idle roughly and stall.

When the vehicle is to be operated continuously below 1,500 m (5,000 feet), turn each pilot screw counterclockwise 1 turn to its original position after removing each pilot screw plug and adjust the idle speed to $1,000 \pm 100$ rpm. Drive new pilot screw plugs into the pilot screw bores (page 4-18). Be sure to do these adjustments at low altitude.

V45 SABRE



V45 MAGNA

